# Graphs of sine and cosine functions (Extended)

## Core Concepts: Graphs of sine and cosine functions (Extended)

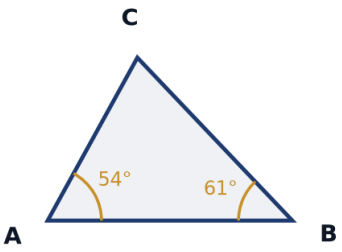
**1** For the model $f(x) = 5x + 6$ in graphs of sine and cosine functions (extended), evaluate $f(2)$.

**a.** 16

**b.** 21

**c.** 10

**d.** 20




**2** Find the positive solutions to the quadratic equation $x^2 - 5x + 4 = 0$.

**a.** $x = 1$ and $x = 5$

**b.** $x = 1$ and $x = 4$

**c.** $x = -1$ and $x = -4$

**d.** $x = 5$ and $x = 4$

**3** A data set recorded for graphs of sine and cosine functions (extended) consists of the four values $\{16, 20, 24, 30\}$. What is the arithmetic mean?

**a.** 20.5

**b.** 21.5

**c.** 22.5

**d.** 24.5

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## Procedural Fluency & Algebraic Manipulation

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**4** For the model $f(x) = 5x + 6$ in graphs of sine and cosine functions (extended), evaluate $f(3)$.

**a.** 21

**b.** 26

**c.** 15

**d.** 25

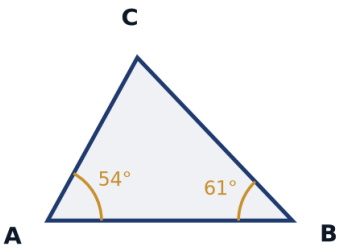
**5** Find the positive solutions to the quadratic equation $x^2 - 7x + 10 = 0$.

**a.** $x = 2$ and $x = 6$

**b.** $x = 2$ and $x = 5$

**c.** $x = -2$ and $x = -5$

**d.** $x = 7$ and $x = 10$




**6** A data set recorded for graphs of sine and cosine functions (extended) consists of the four values $\{16, 20, 24, 30\}$. What is the arithmetic mean?

**a.** 20.5

**b.** 21.5

**c.** 22.5

**d.** 24.5

## Graphical & Geometric Interpretation

**7** For the model $f(x) = 5x + 6$ in graphs of sine and cosine functions (extended), evaluate $f(4)$.

**a.** 26

**b.** 31

**c.** 20

**d.** 30

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**8** Find the positive solutions to the quadratic equation $x^2 - 9x + 18 = 0$.

**a.** $x = 3$ and $x = 7$

**b.** $x = 3$ and $x = 6$

**c.** $x = -3$ and $x = -6$

**d.** $x = 9$ and $x = 18$

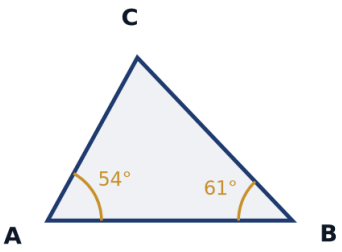
**9** A data set recorded for graphs of sine and cosine functions (extended) consists of the four values $\{16, 20, 24, 30\}$. What is the arithmetic mean?

**a.** 20.5

**b.** 21.5

**c.** 22.5

**d.** 24.5




## Exam-Style Problem Solving & Modeling

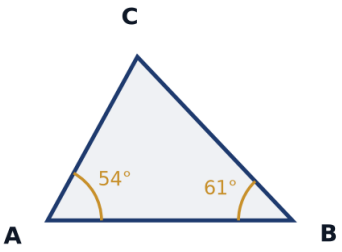
**10** For the model $f(x) = 5x + 6$ in graphs of sine and cosine functions (extended), evaluate $f(5)$.

**a.** 31

**b.** 36

**c.** 25

**d.** 35




**11** Find the positive solutions to the quadratic equation $x^2 - 11x + 28 = 0$.

**a.** $x = 4$ and $x = 8$

**b.** $x = 4$ and $x = 7$

**c.** $x = -4$ and $x = -7$

**d.** $x = 11$ and $x = 28$

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**12** A data set recorded for graphs of sine and cosine functions (extended) consists of the four values $\{16, 20, 24, 30\}$. What is the arithmetic mean?

**a.** $20.5$

**b.** $21.5$

**c.** $22.5$

**d.** $24.5$

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